Table 11.1 Types of compound queries

Compound query	Description
Boolean (`bool`) query	A combination of conditional clauses that wraps individual leaf (term and full-text) queries. Works similarly to the `AND`, `OR`, and `NOT` operators. Example: products= TV AND color = silver NOT rating < 4.5 AND brand = Samsung OR LG
Constant score (`constant_score`) query	Wraps a `filter` query to set constant scores on results. Also helps boost the score. Example: Search for all TVs with user ratings higher than 5 but set a constant score of 5 for each result regardless of the search engine's calculated score.
Function score (`function_score`) query	User-defined functions to assign custom scores to result documents Example: Search for products, and if the product is from LG and is a TV, boost the score by three (via a `script` or `weight` function).
Boosting (`boosting`) query	Boosts the score for positive matches while negating the score for non-matches Example: Fetch all TVs but lower the score of those that are expensive.
Disjunction max (`dis_max`) query	Wraps several queries to search for multiple words in multiple fields (similar to a `multi_match` query) Example: Search for smart TVs in two fields (say, `overview` and `description`) and return the best match.

Of these types, the `bool` query is the most commonly used compound query due to its flexibility and support for multiple conditional clauses. We have a lot to cover when discussing the `bool` query, so it deserves a dedicated section. But before we proceed with the rest of the chapter, we must set up our sandbox to experiment with compound queries.

11.3 The Boolean (bool) query

The Boolean (`bool`) query is the most popular and flexible compound query to create complex criteria for searching data. As the name indicates, it combines Boolean clauses, each with a leaf query made of term-level or full-text queries. Each clause has a typed occurrence of `must`, `must_not`, `should`, or `filter` clauses, briefly explained in table 11.2.

Table 11.2 Boolean clauses

Clause	Description
`must`	An `AND` query where all the documents must match the query criteria Example: Fetch TVs (`product` = `TV`) that fall within a specific price range
`must_not`	A `NOT` query where none of the documents match the query criteria Example: Fetch TVs (`product` = `TV`) that fall within a specific price range *but* with an exception, such as not being a certain color